

CE 310 — Week 2 Weekly Quiz

Week: 2 | **Topic:** Shape Matters: Reading Distributions

Due: Closes **8:00 AM Tuesday, Sep 8** — the day before Exercise 2 and Homework 2 are due. Taken on D2L

Points: 15 questions $\times$ 1 pt each = 15 pts

Attempts: Two — the higher score counts

Time limit: 25 minutes

Instructions: Answer each question in the format it asks for — most are multiple choice, with one matching, one ordering, one select-all-that-apply and one true/false. This quiz is auto-graded and must be submitted before it closes at the time given above. You may use your notes and readings, but work independently.

This quiz covers the Monday lecture and the Wednesday Excel tutorial — histogram bin width, skewness, box-and-whisker plots and the $1.5 \times \text{IQR}$ rule, exploratory data analysis, and the COUNTIFS bin tables that build them in Excel. The tutorial's warning boxes are where the most time-consuming mistakes are flagged, so they are worth reading before you start.

Questions

Q1. In February 2021, over 40 Texas power plants failed during an extreme cold snap, causing blackouts that killed hundreds of people. Power companies had planned capacity based on historical average winter temperatures. Which statistical concept did their planning fail to account for?

- A) The left tail of the winter temperature distribution — rare extreme cold
 - B) The standard deviation of summer peak-demand temperatures
 - C) The median winter temperature across the historical record
 - D) The correlation between air temperature and wind chill at the generating plants
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Q2. A histogram has its tallest bar (mode) on the right side, with a long tail stretching toward lower values on the left. This distribution is best described as:

- A) Right-skewed (positively skewed)
 - B) Left-skewed (negatively skewed)
 - C) Symmetric with zero skewness
 - D) Bimodal
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Q3. An HVAC engineer sizes the dehumidification system for a Tucson office building using the annual average relative humidity (~33%). The 90th percentile of hourly humidity is 64%. What is the most likely design consequence?

- A) The system will be oversized and waste energy year-round
 - B) The system will be correctly sized — annual average is always the appropriate design basis
 - C) The system will automatically switch to a recirculation mode during the monsoon
 - D) The system will be undersized, unable to maintain comfort during peak humidity months
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Q4. [True/False] Changing the bin width of a histogram can change the shape you perceive in the data, so bin width is a choice the analyst has to justify rather than a neutral display setting.

Q5. [Ordering] A dataset has $Q1 = 58^{\circ}\text{F}$ and $Q3 = 86^{\circ}\text{F}$. Put the steps of the $1.5 \times \text{IQR}$ rule in the order you would carry them out to decide which points are outliers.

- (A) Multiply the IQR by 1.5
 - (B) Subtract $Q1$ from $Q3$ to get the IQR
 - (C) Flag any observation lying beyond those fences as an outlier
 - (D) Add that product to $Q3$ and subtract it from $Q1$ to get the upper and lower fences
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Q6. A dataset of annual wind speeds has a skewness coefficient of +1.07. Which statement is correct?

- A) The distribution is left-skewed, and the mean is pulled below the median by rare very-low-wind-speed events
 - B) The distribution is symmetric, so mean = median
 - C) The distribution is right-skewed — most hours have low wind speeds, with a long tail of occasional high-wind hours
 - D) The distribution is right-skewed, but skewness alone provides no information about the relative position of the mean and median
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Q7. Lecture calls exploratory data analysis a conversation with your data before you commit to an analysis, and says skipping it is the most common reason a model fits its training data and then fails in the field. Why does it belong before the fitting rather than after?

- A) It is where the modelling decisions get made — what to transform, which outliers are genuine

- B) It is a formal statistical test whose result determines whether a model may be fitted at all
 - C) It documents the dataset for reviewers, so it belongs with the write-up rather than the analysis
 - D) It reduces the dataset to the rows a model will accept, which has to happen before any fitting
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Q8. A colleague runs `=SKEW()` on a year of daily building electricity use, gets **0.02**, and concludes the data is symmetric and well summarized by its mean. The histogram shows two distinct humps — one near 400 kWh, one near 1,200 kWh — with a valley between them. What is the correct reading?

- A) Skewness near zero confirms the mean is a good summary; the two humps are noise
 - B) The distribution is left-skewed, and the coefficient was computed on the wrong column
 - C) The data mixes two populations, so the mean falls in the valley and describes neither
 - D) The bins are too narrow; widening them would resolve the humps into a single peak
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Q9. [Matching] Match each Excel function to what it returns.

Choices:

- (A) `=STDEV(range)`
- (B) `=PERCENTILE(range, 0.95)`
- (C) `=SKEW(range)`
- (D) `=QUARTILE(range, 1)`
- (E) `=MEDIAN(range)`

Matches:

1. The middle value of the range
 2. The value below which 95% of observations fall
 3. A coefficient describing the direction and degree of asymmetry
 4. The 25th percentile of the range
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Q10. [Select all that apply] Which of the following are part of the five-number summary that a box-and-whisker plot displays?

- (A) The minimum
- (B) The mean
- (C) The first quartile (Q1)
- (D) The median
- (E) The standard deviation

Q11. Your bin table has bin boundaries in column A and you are filling the COUNTIFS formula down all the bins at once. The data range is written Data!B\$2:B\$8761 but the boundary cell as plain A7. Why the difference?

- A) Both should be locked; the missing \$ on the boundary cell is a typo
- B) The data range must not move as the formula fills down, but the boundary must
- C) The \$ makes Excel recalculate the range faster across all 8,760 rows of data
- D) The \$ marks the range as text, which is what COUNTIFS needs for a comparison

Q12. Which formula counts the hours in the 60–70 °F bin, counting 60 itself but not 70?

- A) =COUNTIFS(B2:B8761, ">=60", B2:B8761, "<70")
- B) =COUNTIFS(B2:B8761, ">60", B2:B8761, "<=70")
- C) =COUNTIF(B2:B8761, ">=60") – COUNTIF(B2:B8761, ">=70") is the only way, as COUNTIFS cannot take one range twice
- D) =COUNTIFS(B2:B8761, ">=60", "<70")

Q13. Your box-and-whisker chart of the 8,760 hourly temperatures has the box running from 58 °F to 86 °F, the median line at 73 °F, and whiskers reaching 28 °F and 112 °F. What does the **box** itself tell you?

- A) 58 °F and 86 °F are the coolest and warmest hours recorded during the year
- B) The hours are spread evenly between 58 °F and 86 °F, at a steady rate per degree
- C) Temperatures stayed between 58 °F and 86 °F except during the outlier hours shown as dots
- D) Half of the year's hours fall between 58 °F and 86 °F, and half fall outside that band

Q14. You type =COUNTIFS(Data!B\$2:B\$8761, ">=A7", Data!B\$2:B\$8761, "<(A7+10)") for your bin table and every bin returns 0. Writing the criteria as ">="&A7 and "<"&(A7+10) instead fixes it. What was wrong with your version?

- A) COUNTIFS cannot take a cell reference in a criterion at all — boundaries must be typed as numbers
- B) The data range needs \$ on the column letters too, or the criteria are read against the wrong column

- C) Quoting a cell reference compares the data against the literal text A7, not the number in cell A7
 - D) Filling the formula down shifts A7 by a row each time, so only the first bin can return a count
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Q15. Your histogram is built from the COUNTIFS bin table, but the bars still have gaps between them. Setting **Gap Width to 0%** closes them. Why does that matter?

- A) A gap above zero makes Excel recount the bins, so the bar heights come out wrong
- B) Bars span adjacent ranges, so a gap turns one distribution into separate categories
- C) Gap Width must be 0% or Excel refuses to label the horizontal axis with the bin boundaries
- D) The gap is purely cosmetic; 0% is a house style and does not change how the chart is read